Comprehensive Project Report & Rubric Mapping

Course: DBMS Mini Project (UE24CS395) • **Institution:** PES University (2026–2027)
Team: Saatvik Gupta (PES1UG24CS395) & Satyam Goyal (PES1UG24CS424)

1. Executive Summary & Architecture

The **Intelligence Fusion Platform** is a polyglot investigation management system uniting a 22-table normalized PostgreSQL relational core (with zero-config SQLite failover) with an elastic MongoDB document store. Built with FastAPI and Next.js 15, the platform enables cross-database federated search, relationship network graph analysis, and 64-dimensional semantic vector matching.

2. Official 14-Component Rubric to Evidence Mapping Table

#	Evaluation Component	Marks	Primary Submission PDF / Artifact	Status
1	ER Diagram	2	01_ER_Diagram.pdf (docs/er-diagram.md)	Complete
2	Relational Schema, Keys & Constraints	3	02_Relational_Schema.pdf (docs/database-design.md)	Complete
3	SQL Implementation (DDL & Constraints)	4	03_SQL_DDL_Implementation.pdf (database/sql/schema.sql)	Complete
4	NoSQL / Vector Database Creation	4	04_NoSQL_Vector_Database_Creation.pdf	Complete
5	Technology Justification	1	05_Technology_Justification.pdf	Complete
6	DML Operations & Transactions	4	06_DML_Operations.pdf (database/sql/dml-examples.sql)	Complete
7	Nested, Correlated, Recursive & Joins	4	07_Queries_Nested_Correlated_Recursive_Joins.pdf	Complete
8	Advanced SQL (Procedure, Trigger, CTE)	6	08_Advanced_SQL_Procedure_Function_Trigger_CTE.pdf	Complete
9	NoSQL Feature, Vector Search & RUD	7	09_NoSQL_Vector_Features_and_RUD.pdf	Complete
10	Next.js Interface (TypeScript)	3	frontend/ (18/18 static pages generate cleanly)	Complete
11	Backend API & Database Integration	2	backend/app/ (FastAPI, Swagger, Federated Search)	Complete
12	Authentication & Database RBAC	4	10_Authentication_and_RBAC.pdf (scripts/demonstrate_db_rbac.py)	Complete
13	Documentation, Git & GitHub	2	README.md, .gitignore, clean root repository	Complete
14	Originality, Contributions & Viva	4	docs/modifications.md, docs/viva-preparation.md	Complete
TOTAL	**Official University Rubric Evaluation**	**50**	**All 14 Components Covered with Concrete Code & Output**	**100% Ready**

3. Team Roles & Contribution Summary

• **Saatvik Gupta (PES1UG24CS395):** Relational data modeling (3NF/BCNF DDL/DML), PostgreSQL procedures, triggers and CTEs, FastAPI backend architecture, and Next.js frontend UI overhaul.
• **Satyam Goyal (PES1UG24CS424):** NoSQL document modeling, MongoDB schema validation, RBAC privilege specifications (roles-and-grants.sql), testing scenarios, and academic submission documentation.